PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Effects of integrated models of care for diabetes and hypertension in low-and middle-income countries. A systematic review and meta-analysis
AUTHORS	Rohwer, Anke; Nicol, Jeannine; Toews, Ingrid; Young, Taryn; Bavuma, Charlotte; Meerpohl, Joerg

VERSION 1 – REVIEW

REVIEWER	Berhe, Derbew Fikadu Mekelle University, School of Pharmacy, Mekelle University
REVIEW RETURNED	09-Sep-2020

GENERAL COMMENTS	**1. General comment:** **I read the manuscript with interest. The paper has an important topic (integrated models of care for diabetes and hypertension). My major concern is the statistical analysis and main conclusion. As this study included only few papers, it would be good to describe the data as the topic (integrated models of care) is very complex issue and require in-depth understanding and context analysis. The countries where included papers conducted may have different healthcare set up and system. The ineffectiveness of integrated models of care in these countries should not lead to generalization. The outcome measure used was uncertainty level (high to very low) and finding showed ''very uncertain''. I argue this term should not be used on discuss and conclusion part as it may mislead readers. With intelligent guess, integrated care mode should bring be more effective than the usual care. Therefore, it would be good to address possible limitations of the study and/or problems with the care model.** **There are so many personalized (''we'') sentences. Almost all sentences begin with ''we''. It is not bad to use but not this much. This applies for abstract, method, and discussion part too.**

	2. The Strengths and limitations of this study part (page 2) 2.1 It looks method section as it describes all that has been done on this paper. 2.2 All sentence started with ''we''. May be good to rewrite 2.3 Study limitation not mentioned at all/ contradicting the title given to this section **3. Abstract** **The abstract is well written though with some comments.** 3.1 **What data bases were used should be mentioned on the method part.** 3.2 **If it does not exceed the word limit, it would be nice to explain more the outcome measure.** 3.3 **On the result section, it is good to mention from how many retrieved papers they end up with five.** 3.4 **The PROSPERO registration should be included on method section (unless it is BMJ open format)** 3.5 **What is the total sample size of the included studies?** 3.6 **On conclusion part, I would suggest to rewrite the 1st sentence eg According to the** current evidence, effects of integrated care on health outcomes is very uncertain. 3.7 **On conclusion,** what is ''very uncertain'' mean? This should be written in a clear language. 3.8 If it is ''uncertain'', what did they suggest? At least one suggestion point/the way forward should be added. 3.9 The conclusion given ''very uncertain'' is too much for this paper as they have very few papers. The study has very tight inclusion criteria (study type) and found only five studies. Authors could have assessed this at the early stage of the study and include all types of study (including cross-sectional studies) and then they can conclude the uncertainty. As described on page 6, outcome is described as certainty of evidence (high, moderate, low, or very low). Methodologically this is ok but when discussing the main finding and conclude output, these terms should not be used as such as it may create confusion.

	If the main conclusion is ‘’uncertainty’’, why would readers go deep into this paper. I suggest authors understand their main finding explain it in a clear language. On page 4 (line 4-16) described the effectiveness of integrated care (though descriptive). This implies there is at least evidence to support the integrated care and not appropriate to conclude the effectiveness is ‘’uncertain’’ **4. Introduction part** 4.1 **Why DM and hypertension are chosen? And why only HIV? All CDs also need integrated care. Had it been done for broader NCD and CD, authors could have found more articles. No clear justification is given why on PHC. Why not on hospitals?** 4.2 **Overall, the instruction part gives the prevalence of various diseases and lacks a strong argument on the importance of this study.** **5. Method** **5.1 No clear justification is given why only on PHC.** **5.2 Authors have extensive search strategy. On page 33/34, how were items on #25 (**#21 OR #22 OR #23 OR #25) organized? i.e how are countries #21, #22, #23 and #24 categorized? **5.3** On search strategy: where are the strategies for other databases? 5.4 **On page 33, item #9, it would have been nice to have multidisciplinary team (MDT) as search term.** 6. **Results** 6.1 **There are so many personalized (‘’we’’) sentences. Almost all sentences begin with ‘’we’’. It is not bad to use but not this much. This applies for abstract, method, and discussion part too.** 6.2 **For all tables with reference cited on 1st column: would it be good to add country Author/Publication year [ref]/Country** 6.3 **Table 1: Would it be possible to include what are the main finding/conclusion of each included paper?** 6.4 **For all tables, I would remove all vertical lines** 6.5 Table 2: List of excluded studies. I would make this table as a supplement

	6.6 Table 3: I would split column one of this table 6.7 I would suggest making Table 4 as a supplement and the forest plots supplementary file 5 as the main figure in the main text. (Switching) 6.8 (Table 5) and forest plots are available in supplementary file 4. I am not sure if this can be done on two papers. 6.9 Overall, the result section seems to have quantitative data (meta-analysis) despite the title said it is a systematic review. I would recommend, to remove the statistically analysis and narrate/describe the data as it was done on a few papers. 7. **Discussion** **7.1** Authors should rephrase the idea on **page 20 line 11-14** '' Result suggest that integrated models of care compared to usual care may make little or no difference ……..'' as they have done it on a limited number of papers. Integrated models of care is expected and reported to be effective (also described in the introduction part of this paper). The world is moving towards that direction. Therefore authors should carefully recheck their data, write their findings with their limitations. Or else authors should provide a possible explanation of why it is not effective. For instance, there may be a problem with the implementation, human capacity, or facility issues. **7.2** Authors should provide detailed Strengths and limitations of this study on the discussion part **8.** Conclusion **See comment on abstract Conclusion part.**

REVIEWER	Engelgau, Michael NIH Bethesda
REVIEW RETURNED	27-Oct-2020

GENERAL COMMENTS	Overview The authors conducted a systematic review and metanalysis of integrated care for NCDs and HIV in LMICs. They retrieved studies up to 2019 from several sources. They find 6 manuscripts from 5 different study that are included in their final analyses. They compare integrated care to usual care in rural LMICs setting among patients with various combinations of morbidities (diabetes, hypertension, HIV). They report that integrated care systems adds little benefit to usual care in these settings. The also find that the level of evidence is very uncertain across all their findings. General Comments

	This is an important topic in LMICs and the authors should be commended for this work. This study has a clear methodology which was published prior to this manuscript (the authors cite this publication). There are some areas where the manuscript could be improved. These include: better description of the literature search timeframe; better descriptions of what integrated care includes and how it varies from usual care/stand-alone care, with various terms being used in the text; better understanding of interventions' implementation to help interpret the main findings; better understanding how all of the included studies being done in rural areas may have affected results – in the discussion the authors cite an evaluation of ICDM care quality that found that PHC facilities experienced BP drug stock-outs, lack of functioning BP machines and staff shortages and other challenges which impacted on the delivery of care; and finally more discussion on the point that all the major findings were very uncertain. The literature search timeframe does have an end date but does not seem to have a start date. This needs to be clarified. Integrated care, standard care, and stand-alone care are all used in the intervention descriptions. There needs to be some clear standard definitions for each. The is problematic when we compare table 4 with table 5 and the difference in the interventions is not clear from the titles, nor in the descriptions. The point that all the included studies were done in rural areas is one that needs further examination. Especially in light that the little information we have on implementation finds that these clinics were struggling to deliver integrated care (or at lease anything much better than the "usual" care). The ICDM qualitative study seems to detail this. It may be the case, and likely is, for the other study that occurred in similar settings. This situation would tend to result in a null finding as was the case with this study (i.e., usual care is no better that integrated care – mainly because it is not being delivered). Finally, since essentially all the major findings have large uncertainty, we need to understand how this information is helping create new knowledge in the area of health care delivery. The limited number of included studies, the high levels of potential bias, and the persistent high level of uncertainty around the findings suggest that this type of systematic review and metanalysis may not have enough quality literature to allow for it to be conducted at this point. The should be addressed by the authors in the discussion along with the best way to use the data and interpret results with so few high quality studies that can be included. Specific Comments • The title could be changed to reflect the more broad mix of morbidities include in the studies retrieved. • The introduction could be shorted by ½ to 2/3rds. Brief statements about the disease burden in LMICs are sufficient. Then the study rationale and purpose can be better highlighted. • The World Bank definition for LMICs need to have a date associated with it. It does change over time. • The methods has secondary outcomes listed but few are included in the studies. Suggest to trim this since it does not add much. • More detail is need on the role of the research assistance who screened titles and an understanding why this person was not a

	coauthor, etc. • Tables 4 and 5 seem to be very similar and would suggest to combine. In the discussion, the authors explain these 2 tables' comparisons as "integration of chronic disease services compared to usual care" vs "interventions promoting integrated care compared to usual care" – these seem very similar and why they are separated is unclear.

REVIEWER	Treadwell, Jonathan ECRI, Technology Assessment Group
REVIEW RETURNED	15-Feb-2021

GENERAL COMMENTS	Major comments In GRADE, a key piece of input for each outcome is the MID: the minimal important difference. See background in the 2005 paper by Holger and Gordon in Health Services Research 40:2 (April 2005). An MID is important not only for continuous outcomes such as BP, but also for risk ratios. If we have a risk ratio of say 1.5, then we likely would have a statement that there may be an effect, regardless of the confidence interval (see GRADE guidance by Santesso in Journal of Clinical Epidemiology 119 (2020) 126e135). However if the risk ratio is only 1.09, then we likely would have a statement that there is no important effect, again regardless of the confidence interval. Somewhere in between those extremes of 1.09 and 1.45 is an MID, which is the point at which you go from saying one thing to saying a different thing. With your data, for example, for comparison 1 mortality you had an effect of RR=0.9 for which the converse is 1.11 and you said "little or no effect". Whereas for comparison 1 bp and hiv control you had RR of 1.22 or 1.24 and you said "may increase". So it seems like you were implicitly using an MID of about 1.2? Keep in mind that with GRADE guidance by Santesso, the confidence interval is only about precision, and should not solely decide whether you can make a statement (I say this because the only case where you suggest there may be an increase was also the only case that you had $p<0.05$, which was control of both BP and HIV). Be aware that it is possible in GRADE to have n.s. and still make a statement that there may be an effect (eg point estimate of RR 1.5) since that's your best guess as to where the effect is. You were concerned about performance bias in the three RCTs due to a lack of blinding, however you were not concerned about the lack of blinding in the two ITS studies (the item does not even appear for the two ITS studies). This is inconsistent; either knowledge of treatment group matters, or it does not matter. Since with ITS it appears impossible to blind, I would think you would downgrade them for study limitations on this basis. There may be additional risk of bias domains that you should also reconsider. Minor comments (no change necessary) Generally very well written In the abstract, instead of "Four studies conducted in Sub- Saharan Africa and one in India" consider more specifically saying "Three studies were conducted in South Africa, one was in Uganda/Kenya, and one was in India." The abstract says "risk of bias was moderate", but this is only one of the domains of GRADE…the abstract should not present only a single domain. What you should do is, for each conclusion that you

	draw, provide the GRADE for the evidence on which the conclusion is based. For example you say integrated models “may make little or no difference to mortality” and alongside it you should state the corresponding GRADE (Very Low). Or if that feels cumbersome, simply state that the evidence for all statements had a GRADE of Very Low. In GRADE the domain name is “Study limitations” not “risk of bias”. Please change throughout. Bottom of page 3, to that sentence, when you say “a few”, at the end of that sentence please cite all reviews that you’re talking about. The methods sections says “We defined LMICs according to the World Bank”. Please provide here how the World Bank defines this, so that the reader can judge the appropriateness of their definition. I think you need more text explaining why you are distinguished between the ACTUAL EFFECT of integrating care vs the effect of PROMOTING integrated care. Just a few sentences, perhaps near the beginning of the results section. Without this, some readers may wonder why you didn’t combine all 5 studies into the same meta-analysis. (no change necessary) GRADE table is good and transparent Wording of conclusions in the rightmost column of the GRADE table conforms to GRADE guidance (Santesso et al. / Journal of Clinical Epidemiology 119 (2020) 126e135). Authors should cite that paper and the methods section should note that they followed that guidance (nice job). (no change necessary) Authors appropriately required that in order to qualify as an “interrupted time series”, there had to be at least three datapoints BEFORE the intervention. Please note explicitly that none of the studies reporting rates of diabetes complications such as foot problems or eye problems. (no change necessary) table 3 is excellent thank you for including it In Table 4 and Table 5, in the Effect column, you often simply describe the results narratively, but you do have enough space to provide the numbers, so I’d say do that. One problem with your current approach is that when you say “the RCT found no effect”, does that just mean it was not statistically significant? But perhaps the point estimate was not trivial. I^2, while popular, is only an indirect measure of heterogeneity. See the article by Rucker in BMC Med Res Methodol 8(1) p79. The problem with I2 is that it depends on the Ns. Try it yourself: take a meta-analysis and look at I2. Now triple the Ns, but keep all the effect sizes the same. See how I2 increased? Yeah. That’s not good. Note that tau would not have increased, since it more purely measures the extent to which effect sizes differed. With a large meta-analysis such as yours, even small between-study differences can result in a large I^2, but they won’t give a large tau. Tau may actually make your results seem less heterogeneous. Recognize that tau is on the scale of the effect size metric, so you can use that metric’s MID to decide whether heterogeneity is important.

	Your methods section should state how you decided on a final GRADE when you had both RCTs and ITS in the same outcome (eg number of people achieving BP control). For that outcome did you first come up with two GRADEs (one for each design) and then choose the lower one as the overall GRADE for that outcome? Methods section says “No patients were involved in this systematic review” but technically none were involved in the DEVELOPMENT of this systematic review. They were involved in the studies themselves obviously. (no change necessary) numbers in the PRISMA diagram are consistent Remind the reader in the methods section that non-randomized studies in GRADE start at Low. “excluded 37 articles with reasons (Table 2)” you likely meant “excluded 37 articles (see reasons in Table 2) In a supplementary table, please state for each study at baseline the # who had diabetes and the # who had hypertension and if there were any patients who had neither diabetes nor hypertension. For example the 878 enrolled in Ameh 2017, it would be nice to know these three numbers. For Fairall 2016, patients had “one or more of the following: hypertension, diabetes, chronic respiratory disease, depression”. Do you mean TWO or more? If they only had ONE of these, wouldn’t those single-condition patients be outside the scope of your review? I’d be interested to see how many only had one of the conditions. Discussion, the 2nd paragraph seems unnecessary, since you already said that information in the Methods

REVIEWER	Whitehouse, Christina Villanova University, College of Nursing
REVIEW RETURNED	04-Mar-2021

GENERAL COMMENTS	This is a SR of the effects of integrated models of care for patients living with diabetes and hypertension in low and middle-income countries. Below please see my comments/suggestions for your consideration: 1. Background – unclear why discussion of HIV – the focus of the SR is for DM and HTN – if HIV was also a focus this should be in the search string and in your aims/purpose 2. Not clear to me why this SR is needed, you cite Haldane’s SR which seems similar – where is the gap? Additionally, other reviews that seem similar are indicated – please be clear on how your study stands out among them. What new question do you have that would warrant this review? 3. Methods: a. a. p 6 ln 33 missing “were”. b. Please clarify the inclusion of studies. Definition of partial integration of services is not clear – currently the way written appears to include any/all illness/disease. Your definitions of partial and full integration of services appear very similar – please provide clear definitions c. redundant sentence P 6 ln 40 – you speak of this in the next sentence – consider removing.

	d. I recommend presenting this section using a PICO format. This would help readers understand/clarify what is being compared and evaluated. e. P 6 ln 51 – what does “as defined and reported by primary study authors” mean? Doesn’t appear to be necessary if studies reported the primary and secondary outcomes. f. P6 ln 52 – indicates primary and secondary outcomes – were the outcomes listed a target of your search OR was this what your findings were – unclear. If it was a target then should state at inclusion and/or exclusion criteria. If this is findings they should not be reported here. 4. Results: a. P8ln35 – indicate you classified one study as “awaiting assessment” will you then add an addendum to this report. Unclear of what this means – are you saying your review is ongoing? b. Table 1 – clearly identifies the characteristics of each study. Very useful and concise layout. Prabha koran, 2019 – missing word “patients confirmed diagnosis of ? or hypertension” c. Table 2 - not sure if this is necessary? 5. Conclusion a. Nice summary of findings and comparison to other literature. Discussion is heavily focused on the HIV population considering your findings this is vital as this review is not limited to DM and HTN. Would consider changing the paper title to include HIV. b. Where are your study limitations noted? PRISMA states pg 19 – none noted 6. Language - Change term “diabetic” to “persons with diabetes” Limitations – did not include studies with more than one treatment arm

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1	
1. I read the manuscript with interest. The paper has an important topic (integrated models of care for diabetes and hypertension). My major concern is the statistical analysis and main conclusion. As this study included only few papers, it would be good to describe the data as the topic (integrated models of care) is very complex issue and require in-depth understanding and context analysis. The countries where included papers conducted may have different healthcare set up and system. The ineffectiveness of integrated models of care in these countries should not lead to generalization. The outcome measure used was uncertainty level (high to very low) and finding showed ‘’very uncertain’’. I argue this term should not be used on discuss and conclusion part as it may mislead readers. With intelligent guess, integrated care mode	We conducted a systematic review of available studies assessing the effectiveness of integrated models of care for hypertension and diabetes. We conducted a comprehensive search and reported our methods linked to eligibility of included studies. We used the GRADE approach to assess the certainty of evidence. This approach is widely used as part of systematic reviews and guideline development. It assesses the certainty of evidence per outcome across included studies, taking into account study limitations, inconsistency, imprecision, indirectness and publication bias when downgrading evidence from high to very low certainty. We report this in our methods section. We formulated our conclusions based on guidance provided by the GRADE working group, [1] which we have now referenced and explained in our methods section. When we say the evidence is uncertain, it does not imply that the intervention is ineffective, rather that we don’t know whether it is effective, ineffective, or makes no difference – we report this in our discussion and

should bring be more effective than the usual care. Therefore, it would be good to address possible limitations of the study and/or problems with the care model.	conclusion. We discuss the limitations of our study in the Discussion section and highlight that there are other questions that need to be addressed (other than effectiveness) when implementing integrated care.
2. The Strengths and limitations of this study part (page 2) 2.1 It looks method section as it describes all that has been done on this paper.	The sentences referring to the methods are listed as part of the strengths of our review.
2.2 All sentence started with ‘’we’’. May be good to rewrite	We used the active voice throughout the paper and thus did not make any changes to these sentences.
2.3 Study limitation not mentioned at all/ contradicting the title given to this section	The last bullet point lists a limitation of our review: “Our review did not aim to answer questions on aspects linked to implementation of integrated models of care and barriers and facilitators to integrated models of care at individual and health-system level”
3. Abstract The abstract is well written though with some comments. 3.1 What data bases were used should be mentioned on the method part.	We have added the databases that were searched to the methods section of the abstract.
3.2 If it does not exceed the word limit, it would be nice to explain more the outcome measure.	The word limit does not allow us to expand on the outcome measurement in the abstract.
3.3 On the result section, it is good to mention from how many retrieved papers they end up with five.	We revised the sentence as follows: “We screened 7568 records and included five studies.”
3.4 The PROSPERO registration should be included on method section (unless it is BMJ open format)	The PROSPERO registration number is listed at the end of the abstract.
3.5 What is the total sample size of the included studies?	One of the included studies did not report the sample size and we have thus opted not to provide a total sample size, as this would not be a true reflection of the sample size.
3.6 On conclusion part, I would suggest to rewrite the 1st sentence eg According to the current evidence, effects of integrated care on health outcomes is very uncertain.	Thank you for the suggestion. We have not modified this sentence.
3.7 On conclusion, what is ‘’very uncertain’’ mean? This should be written in a clear language.	In the conclusion of the main text, we explain this: “The evidence on the effectiveness of integrated models of care for people with diabetes, hypertension and other co-morbidities, on health outcomes is very uncertain. We therefore do not know whether integrated models of care lead to better or worse outcomes, or may make no difference at all among people with diabetes, hypertension and other chronic conditions.” The word limit unfortunately does not allow us to explain this in more detail in the abstract.
3.8 If it is ‘’uncertain’’, what did they suggest? At least one suggestion point/the way forward should be added.	We provide this as part of the conclusion: “Programmes and policies on integrated care must consider context-specific factors related to health systems and populations.”

3.9 The conclusion given ‘’very uncertain’’ is too much for this paper as they have very few papers. The study has very tight inclusion criteria (study type) and found only five studies. Authors could have assessed this at the early stage of the study and include all types of study (including cross-sectional studies) and then they can conclude the uncertainty. As described on page 6, outcome is described as certainty of evidence (high, moderate, low, or very low). Methodologically this is ok but when discussing the main finding and conclude output, these terms should not be used as such as it may create confusion. If the main conclusion is ‘’uncertainty’’, why would readers go deep into this paper. I suggest authors understand their main finding explain it in a clear language. On page 4 (line 4-16) described the effectiveness of integrated care (though descriptive). This implies there is at least evidence to support the integrated care and not appropriate to conclude the effectiveness is ‘’uncertain’’	Please see response to comment 1 above. We deliberately included study designs that are able to provide reliable evidence on the effects of integrated models of care on health and process outcomes, according to the Cochrane Effective Practice of Care (EPOC) group. We feel that other study designs that are less suited to answer questions on effectiveness of interventions will not change our results. We have added some further information on this in the strengths and limitation section of the discussion.
4. Introduction part 4.1 Why DM and hypertension are chosen? And why only HIV? All CDs also need integrated care. Had it been done for broader NCD and CD, authors could have found more articles. No clear justification is given why on PHC. Why not on hospitals?	We included hypertension and diabetes as the burden of these two NCDs is rapidly rising in low-and middle-income countries. As per our inclusion criteria, we included participants with hypertension and diabetes plus any other disease (not only HIV). However, the burden of HIV is extremely high in sub-Saharan Africa and other LMICs, and thus a very relevant comorbidity, as explained in the introduction. Furthermore, a lot of changes have been made to health systems to handle the increasing burden of HIV and there is thus an opportunity to leverage these well-functioning systems to provide care for NCDs. We address this in the discussion. We were interested in primary health care settings, as these provide routine care for chronic conditions in LMICs and are usually the first point of contact with the health system for people living in LMICs. Generally, people are only referred to higher levels of care when complications arise. We were not interested in acute treatment and/or complications. We have revised our background and hope that this is clearer now.
4.2 Overall, the instruction part gives the prevalence of various diseases and lacks a strong argument on the importance of this study.	We provide a rationale for conducting the systematic review in the last paragraph of our introduction. There is a lack of reliable evidence on the effectiveness of integrated care in LMICs and thus a thorough assessment of available evidence is warranted.
5. Method **5.1** No clear justification is given why only on PHC.	See response to comment 4.
5.2 Authors have extensive search strategy. On page 33/34, how were items on **#25 (**#21 OR #22 OR #23 OR #25) organized? i.e how are countries #21, #22, #23 and #24 categorized?	This reflects the standard search string for LMICs. Our search was done by an information specialist and we are confident that it was well executed.
5.3 On search strategy: where are the strategies	We have added the search strategies for all

for other databases?	electronic databases to Supplementary file 1.
5.4 On page 33, item #9, it would have been nice to have multidisciplinary team (MDT) as search term.	The search string #9 refers to multi-morbidity. 'multi-disciplinary team' refers to a different term.
Results 6.1 There are so many personalized (''we'') sentences. Almost all sentences begin with ''we''. It is not bad to use but not this much. This applies for abstract, method, and discussion part too.	We wrote our manuscript using the active voice, which is standard practice.
6.2 For all tables with reference cited on 1st column: would it be good to add country **Author/Publication year [ref]/Country**	Table 1 includes a column on 'Country/Setting'. We prefer to leave the study ID as Author/Year of publication.
6.3 Table 1: Would it be possible to include what are the main finding/conclusion of each included paper?	Table 1 is a summary of the characteristics of included studies. The findings are presented in the sections: Integration of chronic disease services compared to usual care; and Interventions promoting integrated care compared to usual care.
6.4 For all tables, I would remove all vertical lines	Thanks for you comment. We prefer to include vertical lines in the tables and assume that BMJ Open will format them according to their preference when publishing the paper.
6.5 Table 2: List of excluded studies. I would make this table as a supplement	We have moved the table of excluded studies to Supplementary file 2.
6.6 Table 3: I would split column one of this table	Thanks for the suggestion. We prefer to keep the Name and the Study ID in one column.
6.7 I would suggest making Table 4 as a supplement and the forest plots supplementary file 5 as the main figure in the main text. (Switching)	Tables 4 and 5 are the Summary of findings table and therefore the most important tables linked to the findings of our review. The forest plots only depict the pooled effect and not the comprehensive findings and GRADE assessments. We were also not able to pool data for all outcomes.
6.8 (Table 5) and forest plots are available in supplementary file 4. I am not sure if this can be done on two papers.	
6.9 Overall, the result section seems to have quantitative data (meta-analysis) despite the title said it is a systematic review. I would recommend, to remove the statistically analysis and narrate/describe the data as it was done on a few papers.	We report in our methods section that we performed meta-analysis where studies were sufficiently homogenous, and narrative synthesis where heterogeneity was high. Meta-analysis refers to the statistical analysis and not to the type of study that was conducted. We have added the term 'meta-analysis' to our title as per the editor's suggestion.
7. **Discussion** **7.1** Authors should rephrase the idea on **page 20 line 11-14** '' Result suggest that integrated models of care compared to usual care may make little or no difference ……..'' as they have done it on a limited number of papers. Integrated models of care is expected and reported to be effective (also described in the introduction part of this paper). The world is moving towards that direction. Therefore authors should carefully recheck their data, write their findings with their limitations. Or else authors should provide a possible explanation of why it is not effective. For instance, there may be a problem with the implementation, human capacity, or facility issues.	Our conclusions are based on the available evidence. As per previous comments, we conducted a comprehensive search and used systematic methods to select studies, extract and analyse data, and make judgements on the certainty of evidence. We did not conclude that integrated care is ineffective, but rather that we do not know whether it is effective, ineffective or makes no difference on the outcomes we assessed. We also discussed issues related to implementation of integrated care in our discussion section.
7.2 Authors should provide detailed Strengths and limitations of this study on the discussion part	We added a subheading on strengths and limitations to the discussion section.

Reviewer: 2	
1. The authors conducted a systematic review and metanalysis of integrated care for NCDs and HIV in LMICs. They retrieved studies up to 2019 from several sources. They find 6 manuscripts from 5 different study that are included in their final analyses. They compare integrated care to usual care in rural LMICs setting among patients with various combinations of morbidities (diabetes, hypertension, HIV). They report that integrated care systems adds little benefit to usual care in these settings. The also find that the level of evidence is very uncertain across all their findings. This is an important topic in LMICs and the authors should be commended for this work. This study has a clear methodology which was published prior to this manuscript (the authors cite this publication). There are some areas where the manuscript could be improved. These include: better description of the literature search timeframe; better descriptions of what integrated care includes and how it varies from usual care/stand-alone care, with various terms being used in the text; better understanding of interventions' implementation to help interpret the main findings; better understanding how all of the included studies being done in rural areas may have affected results – in the discussion the authors cite an evaluation of ICDM care quality that found that PHC facilities experienced BP drug stock-outs, lack of functioning BP machines and staff shortages and other challenges which impacted on the delivery of care; and finally more discussion on the point that all the major findings were very uncertain.	Thank you for your comments. We have responded to the points you raised below (comment 2-12).
2. The literature search timeframe does have an end date but does not seem to have a start date. This needs to be clarified.	We reported the date of the last search (December 2019), as per the PRISMA reporting guidelines. We did not have any restrictions linked to the date of publication. We now clarify this in the methods section: "We did not have any restrictions related to language, date of publication or publication status."
3. Integrated care, standard care, and stand-alone care are all used in the intervention descriptions. There needs to be some clear standard definitions for each.	Thank you for your comment. Included studies did not adequately report on the degree of integration in the intervention and control groups and we therefore categorised all interventions as 'integrated care' or 'interventions to promote integrated delivery of care' and all comparisons as 'usual care'. We have clarified this in the

	methods section under 'Criteria for considering studies for inclusion': **Types of interventions** Eligible interventions were models of full or partial integration of services at PHC and community level. Full integration of service delivery was defined as models where patients (primarily treated for diabetes, hypertension or any other disease) received the full package of care (prevention, diagnosis and treatment) for diabetes or hypertension and any other chronic disease at the same point of care by one or more healthcare professionals. Partial integration of services was defined as models where patients treated for diabetes, hypertension, or any other chronic disease received part of the package of care (either prevention, diagnosis, or treatment) for another disease (see Figure 1). Partially integrated models of care therefore refer to a lower level of integration compared to fully integrated models of care. For example, with partially integrated care, patients receiving treatment for hypertension could be tested for HIV and referred for treatment; whereas with fully integrated care, patients receiving treatment for hypertension would be tested and treated for HIV during the same clinic visit. Included studies did not provide adequate information for us to categorise interventions as fully integrated models of care or partially integrated models of care and we thus categorised interventions as integrated models of care and interventions that promoted an integrated approach to providing care. **Types of comparisons** We aimed to compare fully integrated models of care to stand-alone care; partially integrated models of care to stand-alone care; and fully integrated models of care to partially integrated models of care. However, for all included studies, comparisons were reported as standard or usual care and authors did not provide an adequate description of what that entailed. Although these seemed to refer to less integrated care, we unable to categorise them as partially integrated models of care or stand-alone care. We therefore compared integrated models of care to usual care, and interventions to promote integrated delivery of care to usual care.
4. The is problematic when we compare table 4 with table 5 and the difference in the interventions is not clear from the titles, nor in the descriptions.	Thank you for your comment. We have clarified the difference between these two types of interventions in the methods section, under 'Criteria for considering studies for inclusion': "Integrated models of care assessed the effect of integration of service delivery i.e. integration

	of two previously separate models of delivery of care into one model of delivery of care, for example integrating HIV services into general PHC services. We distinguished these interventions from interventions that promoted an integrated approach to providing care in PHC facilities. In these cases, services as such were not integrated, but healthcare workers were encouraged to provide holistic patient care, for example through the provision and use of clinical management tools that supported an integrated approach to care."
5. The point that all the included studies were done in rural areas is one that needs further examination. Especially in light that the little information we have on implementation finds that these clinics were struggling to deliver integrated care (or at lease anything much better than the "usual" care). The ICDM qualitative study seems to detail this. It may be the case, and likely is, for the other study that occurred in similar settings. This situation would tend to result in a null finding as was the case with this study (i.e., usual care is no better that integrated care – mainly because it is not being delivered).	Thanks for you comment. Indeed, included studies were mostly conducted in rural areas. We have now added this to the discussion section.
6. Finally, since essentially all the major findings have large uncertainty, we need to understand how this information is helping create new knowledge in the area of health care delivery. The limited number of included studies, the high levels of potential bias, and the persistent high level of uncertainty around the findings suggest that this type of systematic review and metanalysis may not have enough quality literature to allow for it to be conducted at this point. The should be addressed by the authors in the discussion along with the best way to use the data and interpret results with so few high quality studies that can be included.	Thanks for your comment. We conducted a comprehensive search and used systematic methods to select studies, extract and analyse data, and make judgements on the certainty of evidence using the GRADE. Our conclusions are thus based on the current available evidence. We feel this still adds value as it presents the status quo of the available evidence and identifies gaps in the evidence. In the discussion, we highlight that the question on effectiveness is just one question that needs to be taken into consideration when making decisions about implementing integrated care. Other questions such as those related to implementation issues, costs, and patient experiences is outside of the scope of our review. We address implementation issues in the discussion section.
7. The title could be changed to reflect the more broad mix of morbidities include in the studies retrieved.	Our question related to integrated models of care for diabetes and hypertension plus other conditions like HIV, TB, depression etc. As per our inclusion criteria, all participants

	should therefore at least have either diabetes or hypertension. This is reflected in the title. Furthermore, our published protocol has the same title, which is another reason we would like to keep the title as is.
8. The introduction could be shorted by ½ to 2/3rds. Brief statements about the disease burden in LMICs are sufficient. Then the study rationale and purpose can be better highlighted.	We provide a brief background to the burden of NCDs and multimorbidity and the need for integrated models of care. The rationale for the study is reported in the last paragraph of the introduction. We have added the following sentence to point out the lack of evidence on the effectiveness of integrated care: "Although integrated models of care have been advocated widely, and various models and programmes have been implemented and described, there is a lack of evidence on the effectiveness of integrated care compared to other models of care in LMICs."
9. The World Bank definition for LMICs need to have a date associated with it. It does change over time.	We used the 2016 classification of countries and have added this to the definition.
10. The methods has secondary outcomes listed but few are included in the studies. Suggest to trim this since it does not add much.	The secondary outcomes listed in the methods sections are prespecified process outcomes that we were interested in. However, few of these outcomes were measured in the included studies. We feel that this is still an important finding.
11. More detail is need on the role of the research assistance who screened titles and an understanding why this person was not a coauthor, etc.	We use the recommendations put forward by the International Committee of Medical Journal Editors (ICMJE)[2] to determine authorship of papers. According to these, the following four criteria have to be met for authorship: 1. Substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data for the work; AND 2. Drafting the work or revising it critically for important intellectual content; AND 3. Final approval of the version to be published; AND 4. Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. We contracted the research assistant to assist with screening of titles and abstracts only, which is a very small part of the systematic review process. The assistant therefore did not contribute substantially to the work (criteria 1).

12. Tables 4 and 5 seem to be very similar and would suggest to combine. In the discussion, the authors explain these 2 tables' comparisons as "integration of chronic disease services compared to usual care" vs "interventions promoting integrated care compared to usual care" – these seem very similar and why they are separated is unclear.	Thank you for your comment. We have clarified the difference between these two types of interventions in the methods section, under 'Criteria for considering studies for inclusion': "Integrated models of care assessed the effect of integration of service delivery i.e. integration of two previously separate models of delivery of care into one model of delivery of care, for example integrating HIV services into general PHC services. We distinguished these interventions from interventions that promoted an integrated approach to providing care in PHC facilities. In these cases, services as such were not integrated, but healthcare workers were encouraged to provide holistic patient care, for example through the provision and use of clinical management tools that supported an integrated approach to care." Tables 4 and 5 (now tables 3 and 4) present the findings of the two comparisons and can therefore not be combined.
Reviewer: 3	
1. In GRADE, a key piece of input for each outcome is the MID: the minimal important difference. See background in the 2005 paper by Holger and Gordon in Health Services Research 40:2 (April 2005). An MID is important not only for continuous outcomes such as BP, but also for risk ratios. If we have a risk ratio of say 1.5, then we likely would have a statement that there may be an effect, regardless of the confidence interval (see GRADE guidance by Santesso in Journal of Clinical Epidemiology 119 (2020) 126e135). However if the risk ratio is only 1.09, then we likely would have a statement that there is no important effect, again regardless of the confidence interval. Somewhere in between those extremes of 1.09 and 1.45 is an MID, which is the point at which you go from saying one thing to saying a different thing. With your data, for example, for comparison 1 mortality you had an effect of RR=0.9 for which the converse is 1.11 and you said "little or no effect". Whereas for comparison 1 bp and hiv control you had RR of 1.22 or 1.24 and you said "may increase". So it seems like you were implicitly using an MID of about 1.2? Keep in mind that with GRADE guidance by Santesso, the	Thank you for pointing this out. We agree with your comment and had indeed considered MIDs and used these implicitly for RRs. For continuous outcomes, we consulted the literature on what a clinically meaningful change in HbA1C, BP and total cholesterol would be, but did not find a specific answer. We also consulted clinicians with expertise in these areas. We discussed this within the author team. As per the guidance from Santesso and colleagues (2020), we interpreted the findings based on the size of the effect and the certainty of evidence.

	confidence interval is only about precision, and should not solely decide whether you can make a statement (I say this because the only case where you suggest there may be an increase was also the only case that you had p<0.05, which was control of both BP and HIV). Be aware that it is possible in GRADE to have n.s. and still make a statement that there may be an effect (eg point estimate of RR 1.5) since that's your best guess as to where the effect is.	
2.	You were concerned about performance bias in the three RCTs due to a lack of blinding, however you were not concerned about the lack of blinding in the two ITS studies (the item does not even appear for the two ITS studies). This is inconsistent; either knowledge of treatment group matters, or it does not matter. Since with ITS it appears impossible to blind, I would think you would downgrade them for study limitations on this basis. There may be additional risk of bias domains that you should also reconsider.	Thank you for your comment. For ITS studies, we assessed blinding by considering whether "knowledge of the allocated intervention was adequately prevented during the study." (see methods section) We followed EPOC guidance to make risk of bias judgements for studies, and assessed the risk of bias for this domain to be low for both of the ITS studies, as the methods for data collection were similar before and after the intervention, and the intervention did not affect data collection. In Rawat 2018, data was collected from the department of health. In Ameh 2017, patients were recruited in June 2013 and data was collected retrospectively from patient records for the period Jan 2011 to June 2013. As ITS studies are observational studies, they start at 'low certainty' evidence already (this takes into account study limitations). We further downgraded certainty of evidence for outcomes reported in the ITS studies for indirectness. The certainty of evidence linked to outcomes reported in the ITS studies as therefore judged to be 'very low'.
3.	(no change necessary) Generally very well written	Thank you.
4.	In the abstract, instead of "Four studies conducted in Sub- Saharan Africa and one in India" consider more specifically saying "Three studies were conducted in South Africa, one was in Uganda/Kenya, and one was in India."	We have changed this as suggested.
5.	The abstract says "risk of bias was moderate", but this is only one of the domains of GRADE…the abstract should not present only a	We have added the certainty of evidence in brackets for each outcome.

	single domain. What you should do is, for each conclusion that you draw, provide the GRADE for the evidence on which the conclusion is based. For example you say integrated models "may make little or no difference to mortality" and alongside it you should state the corresponding GRADE (Very Low). Or if that feels cumbersome, simply state that the evidence for all statements had a GRADE of Very Low.	
6.	In GRADE the domain name is "Study limitations" not "risk of bias". Please change throughout.	We changed this in Table 4 and 5 (now Table 3 and 4).
7.	Bottom of page 3, to that sentence, when you say "a few", at the end of that sentence please cite all reviews that you're talking about.	We have added the citations of reviews included in our scoping review.
8.	The methods sections says "We defined LMICs according to the World Bank". Please provide here how the World Bank defines this, so that the reader can judge the appropriateness of their definition.	We revised our definition as follows: "We defined LMICs according to the 2016 classification of the World Bank, that defined low-income economies as those with a gross national income (GNI) per capita of $1035 or less, lower middle income economies as those with a GNI per capita of $1006 to $3995, and upper middle economies as those with a GNI per capita of $3956 to $12235."
9.	I think you need more text explaining why you are distinguished between the ACTUAL EFFECT of integrating care vs the effect of PROMOTING integrated care. Just a few sentences, perhaps near the beginning of the results section. Without this, some readers may wonder why you didn't combine all 5 studies into the same meta-analysis.	Thank you for your comment. We have clarified this in the methods section, under 'Criteria for considering studies for inclusion': "Integrated models of care assessed the effect of integration of service delivery i.e. integration of two previously separate models of delivery of care into one model of delivery of care, for example integrating HIV services into general PHC services. We distinguished these interventions from interventions that promoted an integrated approach to providing care in PHC facilities. In these cases, services as such were not integrated, but healthcare workers were encouraged to provide holistic patient care, for example through the provision and use of clinical management tools that supported an integrated approach to care."
10.	(no change necessary) GRADE table is good and transparent	Thank you.
11.	Wording of conclusions in the rightmost column of the GRADE table conforms to GRADE guidance (Santesso et al. / Journal of Clinical Epidemiology 119	Thanks. We have added a sentence and the reference to the paper in the methods section.

(2020) 126e135). Authors should cite that paper and the methods section should note that they followed that guidance (nice job).	
12. (no change necessary) Authors appropriately required that in order to qualify as an “interrupted time series”, there had to be at least three datapoints BEFORE the intervention.	Thank you.
13. Please note explicitly that none of the studies reporting rates of diabetes complications such as foot problems or eye problems.	We agree that these are relevant outcomes related to diabetes. However, we did not pre-specify them as important outcomes for our review.
14. (no change necessary) table 3 is excellent thank you for including it	Thank you.
15. In Table 4 and Table 5, in the Effect column, you often simply describe the results narratively, but you do have enough space to provide the numbers, so I’d say do that. One problem with your current approach is that when you say “the RCT found no effect”, does that just mean it was not statistically significant? But perhaps the point estimate was not trivial.	We have added the data as part of the effect column for tables 4 and 5 (now tables 3 and 4).
16. I^2, while popular, is only an indirect measure of heterogeneity. See the article by Rucker in BMC Med Res Methodol 8(1) p79. The problem with I2 is that it depends on the Ns. Try it yourself: take a meta-analysis and look at I2. Now triple the Ns, but keep all the effect sizes the same. See how I2 increased? Yeah. That’s not good. Note that tau would not have increased, since it more purely measures the extent to which effect sizes differed. With a large meta-analysis such as yours, even small between-study differences can result in a large I^2, but they won’t give a large tau. Tau may actually make your results seem less heterogeneous. Recognize that tau is on the scale of the effect size metric, so you can use that metric’s MID to decide whether heterogeneity is	Thanks for pointing this out. We agree with your comment. We were only able to pool results for the outcomes change in HbA1C and change in systolic BP. Findings from the two studies included in the meta-analysis were very consistent and there was no statistical heterogeneity, I^2 was 0% and Tau^2 was 0.00. We also did not downgrade the certainty of evidence for inconsistency.

important.	
17. Your methods section should state how you decided on a final GRADE when you had both RCTs and ITS in the same outcome (eg number of people achieving BP control). For that outcome did you first come up with two GRADEs (one for each design) and then choose the lower one as the overall GRADE for that outcome?	Thank you for pointing this out. We made GRADE judgements for both RCTs and ITS studies separately. Both were judged as having very low certainty evidence for all outcomes and thus the overall judgement was 'very low'. We have added the following sentence to the methods section: "For each outcome, we described the certainty of evidence as high, moderate, low or very low. For outcomes reported by both RCTs and non-randomised studies, we made separate GRADE judgements for both types of studies. Where we arrived at the same level of certainty of evidence, we summarised this in a single judgement per outcome."
18. Methods section says "No patients were involved in this systematic review" but technically none were involved in the DEVELOPMENT of this systematic review. They were involved in the studies themselves obviously.	We have revised the sentence accordingly.
19. (no change necessary) numbers in the PRISMA diagram are consistent	Thank you.
20. Remind the reader in the methods section that non-randomized studies in GRADE start at Low.	We have added this to the methods section.
21. "excluded 37 articles with reasons (Table 2)" you likely meant "excluded 37 articles (see reasons in Table 2)	We have revised this sentence to state: "We excluded 37 articles and reported reasons for exclusion (Supplementary file 2)."
22. In a supplementary table, please state for each study at baseline the # who had diabetes and the # who had hypertension and if there were any patients who had neither diabetes nor hypertension. For example the 878 enrolled in Ameh 2017, it would be nice to know these three numbers. For Fairall 2016, patients had "one or more of the following: hypertension, diabetes, chronic respiratory disease, depression". Do you mean TWO or more? If they only had ONE of these, wouldn't those single-condition patients be outside the scope of your review? I'd be interested to see how many	Not all studies provided the number of participants with hypertension and/or diabetes. Where reported, we have now added this to the Table 1: Summary of characteristics of included studies.

only had one of the conditions.	
23. Discussion, the 2nd paragraph seems unnecessary, since you already said that information in the Methods	We have moved this paragraph to the new subheading 'Strengths and limitations' in the discussion section.
Reviewer: 4	
1. Background – unclear why discussion of HIV – the focus of the SR is for DM and HTN – if HIV was also a focus this should be in the search string and in your aims/purpose	We included hypertension and diabetes as the burden of these two NCDs is rapidly rising in low-and middle-income countries. As per our inclusion criteria, we included participants with hypertension and diabetes plus any other disease (not only HIV). However, the burden of HIV is extremely high in sub-Saharan Africa and other LMICs, and thus a very relevant comorbidity. Furthermore, a lot of changes have been made to health systems to handle the increasing burden of HIV and there is thus an opportunity to leverage these well-functioning systems to provide care for NCDs.
2. Not clear to me why this SR is needed, you cite Haldane's SR which seems similar – where is the gap? Additionally, other reviews that seem similar are indicated – please be clear on how your study stands out among them. What new question do you have that would warrant this review?	The review by Haldane and colleagues examined existing programmes of integrated healthcare delivery for diabetes, hypertension or CVDs with HIV/AIDS. The included studies are mostly descriptive in nature, describing existing programmes with no thorough evaluation of the effectiveness of these programmes. Furthermore, the date of the last search was October 2015 and thus outf date. Indeed, none of our included studies were included in the review by Haldane and colleagues. We deliberately included study designs that are able to provide reliable evidence on the effects of integrated models of care on health and process outcomes, according to the Cochrane Effective Practice of Care (EPOC) group. We feel that other study designs that are less suited to answer questions on effectiveness of interventions will not change our results. Limitations of other reviews are reported in the last paragraph of the background – reviews did not include studies conducted in LMICs; reviews did not include studies assessing integration of care for NCDs, or integrated care for NCDs and communicable diseases. We therefore feel that our questions does address a gap in the current evidence base.
3. Methods: a. a. p 6 ln 33 missing "were". b. Please clarify the inclusion of studies. Definition of partial integration of services is not clear – currently the way written appears to include	Thank you for your comment. Included studies did not adequately report on the degree of integration in the intervention and control groups and we therefore categorised all interventions as 'integrated care' or 'interventions to promote

any/all illness/disease. Your definitions of partial and full integration of services appear very similar – please provide clear definitions	integrated delivery of care' and all comparisons as 'usual care'. We have clarified this in the methods section under 'Criteria for considering studies for inclusion': **Types of interventions** Eligible interventions were models of full or partial integration of services at PHC and community level. Full integration of service delivery was defined as models where patients (primarily treated for diabetes, hypertension or any other disease) received the full package of care (prevention, diagnosis and treatment) for diabetes or hypertension and any other chronic disease at the same point of care by one or more healthcare professionals. Partial integration of services was defined as models where patients treated for diabetes, hypertension, or any other chronic disease received part of the package of care (either prevention, diagnosis, or treatment) for another disease (see Figure 1). Partially integrated models of care therefore refer to a lower level of integration compared to fully integrated models of care. For example, with partially integrated care, patients receiving treatment for hypertension could be tested for HIV and referred for treatment; whereas with fully integrated care, patients receiving treatment for hypertension would be tested and treated for HIV during the same clinic visit. Included studies did not provide adequate information for us to categorise interventions as fully integrated models of care or partially integrated models of care and we thus categorised interventions as integrated models of care and interventions that promoted an integrated approach to providing care. **Types of comparisons** We aimed to compare fully integrated models of care to stand-alone care; partially integrated models of care to stand-alone care; and fully integrated models of care to partially integrated models of care. However, for all included studies, comparisons were reported as standard or usual care and authors did not provide an adequate description of what that entailed. Although these seemed to refer to less integrated care, we unable to categorise them as partially integrated models of care or stand-alone care. We therefore compared integrated models of care to usual care, and interventions to promote integrated delivery of care to usual care.

c. redundant sentence P 6 ln 40 – you speak of this in the next sentence – consider removing.	Not sure which sentence this refers to.
d. I recommend presenting this section using a PICO format. This would help readers understand/clarify what is being compared and evaluated.	Thanks for your comment. We added subheadings to the methods section and hope that this is clearer now.
e. P 6 ln 51 – what does "as defined and reported by primary study authors" mean? Doesn't appear to be necessary if studies reported the primary and secondary outcomes.	This refers to how outcomes were defined and measured in included studies. We did not pre-specify outcome definition or measurement, but included outcomes regardless of how study authors defined and measured them. For example, two studies reported on depression (reported under our outcome 'disease specific morbidity') but used different tools to score depression – we included both of these, as this was how authors of primary studies defined and measured their outcome.
f. P6 ln 52 – indicates primary and secondary outcomes – were the outcomes listed a target of your search OR was this what your findings were – unclear. If it was a target then should state at inclusion and/or exclusion criteria. If this is findings they should not be reported here	The outcomes listed are the ones we pre-specified as being important for our review. This is now reported under the subheading – Criteria for considering studies for inclusion/Types of outcomes.
4. Results: a. P8ln35 – indicate you classified one study as "awaiting assessment" will you then add an addendum to this report. Unclear of what this means – are you saying your review is ongoing?	We report that: "For one study that met eligibility criteria, we were only able to access the conference abstract. We classified this study as 'awaiting assessment', as we are unable to definitively decide on inclusion or exclusion until we have access to the full report". We contacted the author of this study, but they were not able to provide a report.
b. Table 1 – clearly identifies the characteristics of each study. Very useful and concise layout. Prabha koran, 2019 – missing word "patients confirmed diagnosis of ? or hypertension"	Thanks for pointing this out. We have revised this to: "Patients with confirmed diagnosis of diabetes or hypertension"
c. Table 2 - not sure if this is necessary?	According to systematic review methods and PRISMA reporting guidelines, reasons for excluding studies at full-text screening stage should be reported. We moved this table to Supplementary file 2.
5. Conclusion a. Nice summary of findings and comparison to other literature. Discussion is heavily focused on the HIV population considering your findings this is vital as this review is not limited to DM and HTN. Would consider changing the paper title to include HIV.	We considered participants with multimorbidity that included at least diabetes or hypertension plus any other disease. However, the studies that we found mostly assessed integrated care for diabetes/hypertension and HIV. We comment on this in the discussion section: "Although the literature predominantly highlights the need to integrate NCD and HIV care, integrating mental health services into existing NCD and or HIV services is just as important. Four of the five ongoing studies that we identified examine integration of mental health

	with NCDs.”
b. Where are your study limitations noted? PRISMA states pg 19 – none noted	We have now added a section on ‘Strengths and limitations’ to the discussion
6. Language - Change term “diabetic” to “persons with diabetes”	We have only used the term ‘diabetic’ when referring to the outcomes of Rawat 2018. They defined two of their outcomes as ‘population level new diabetics on treatment’ and ‘clinic level new diabetics on treatment’ and we do not want to change this when specifically referring to these outcomes. In all other instances, we refer to people living with diabetes.
Limitations – did not include studies with more than one treatment arm	We have now added a section on ‘Strengths and limitations’ to the discussion.

[1] Santesso N, Glenton C, Dahm P, et al. GRADE guidelines 26: informative statements to communicate the findings of systematic reviews of interventions. *J Clin Epidemiol* 2020;119:126-35. doi: 10.1016/j.jclinepi.2019.10.014 [published Online First: 2019/11/13]
[2] http://www.icmje.org/recommendations/

VERSION 2 – REVIEW

REVIEWER	Engelgau, Michael NIH Bethesda
REVIEW RETURNED	06-Apr-2021

GENERAL COMMENTS	The authors have responded to my comments and suggestions to my satisfaction. Suggest the paper be accepted

REVIEWER	Treadwell, Jonathan ECRI, Technology Assessment Group
REVIEW RETURNED	08-Apr-2021

GENERAL COMMENTS	Second manuscript review Title: Effects of integrated models of care for diabetes and hypertension in low-and middle-income countries. A systematic review and meta-analysis Reviewer: Jonathan R Treadwel, ECRI, Plymouth Meeting PA Since you did not actually require that patients have multimorbidity, under Eligibility criteria as well as the section Overall completeness and applicability of evidence, you can delete "plus any other disease", . Also under "Types of Participants", you can edit "We included studies with adults and children attending PHC clinics, presenting with diabetes or hypertension plus one or more other chronic diseases (multi-morbidity), " to We included studies with

	adults and children attending PHC clinics, presenting with diabetes or hypertension, and patients may potentially have had additional chronic diseases (multi-morbidity)" in methods you wrote "Included studies did not provide adequate information for us to categorise interventions as fully integrated models of care or partially integrated models of care and we thus categorised interventions as integrated models of care and interventions that promoted integrated delivery of care" and i think it would be slightly clearer to say "Included studies did not provide adequate information for us to categorise interventions as fully integrated models of care or partially integrated models of care and we thus categorised interventions as either 1) integrated models of care or 2) interventions that promoted integrated delivery of care" This edit will slightly better prepare the reader for your later separation of these two categories For heterogeneity measurement, as I stated in my first peer review of this manuscript, I recommend tau instead of i^2 and chi^2, as tau is a direct measure of heterogeneity rather than indirect. But I understand the authors wanted to follow the Cochrane recommendation. The methods sections states "We interpreted the certainty of evidence according to guidance provided by the GRADE working group, which takes into consideration the size of the effect and the certainty of evidence." You seem reluctant to be transparent about the implicit numerical thresholds you used when saying "little or no effect" vs "very small effect" vs "may increase". The GRADE guidance recommends transparency. At the very least please change "We interpreted the certainty of evidence according to guidance..." to "We wrote statements about the evidence (e.g., "little or no effect" vs. "very small effect") according to guidance..." in the new table 3, it seems you made something harder to understand, not easier. Specifically instead of "probability of CD4 count control was 6% greater in intervention clinics compared to control clinics" you merely have the beta with its CI. I doubt readers will know what that beta means, so i'd prefer the earlier version. Maybe keep the old text and add beta. for access to care the table only says "There was no change in trend..." And then to the right "little or no effect". However the left column text did not even provide a point estimate of the effect, thus the reader cannot judge whether it is appropriate to say "little or no effect". The text below does give that, so i think the table needs the numbers too. in the strategies to promote table, for promoting you wrote "The mean change in HbA1c with integrated care was 0.11 % higher (0.2 lower to 0.42 higher)". Please check the accuracy of that. It sounds like on average patients got slightly WORSE? Whereas the usual care group got slightly better. So you'll need something in the relative comparison column, which it seems will at least slightly favor usual care. These are two RCTs so probably they reported enough info for you to calculate a relative effect size. Same point with SPB (promotion group got a little worse but the usual care group got a little better).

	in the strategies to promote table, for changes in total cholesterol, you wrote "The mean change in total cholesterol with usual care was 2.0 mg/dl" but is that an increase or a decrease? I'd also like to see numbers in the relative column, in order to judge the appropriateness of your statement that there was "little or no effect"

VERSION 2 – AUTHOR RESPONSE

Reviewer 2 : Dr. Michael Engelgau, NIH Bethesda	
Comments to the Author: The authors have responded to my comments and suggestions to my satisfaction. Suggest the paper be accepted	Thank you very much for the constructive feedback.
Reviewer 3: Dr. Jonathan Treadwell, ECRI	
1. Since you did not actually require that patients have multi-morbidity, under Eligibility criteria as well as the section Overall completeness and applicability of evidence, you can delete "plus any other disease". Also, under "Types of Participants", you can edit "We included studies with adults and children attending PHC clinics, presenting with diabetes or hypertension plus one or more other chronic diseases (multi-morbidity), " to We included studies with adults and children attending PHC clinics, presenting with diabetes or hypertension, and patients may potentially have had additional chronic diseases (multi-morbidity)"	Thank you for feedback, we have rephrased the sentence as suggested.
2. In methods you wrote "Included studies did not provide adequate information for us to categorise interventions as fully integrated models of care or partially integrated models of care and we thus categorised interventions as integrated models of care and interventions that promoted integrated delivery of care" and i think it would be slightly clearer to say "Included studies did not provide adequate information for us to categorise interventions as fully integrated models of care or partially integrated models of care and we thus categorised interventions as either 1) integrated models of care or 2) interventions that promoted integrated delivery of care" This edit will slightly better prepare the reader for your later separation of these two categories	Thanks. We made this edit.
3. For heterogeneity measurement, as I stated in my first peer review of this	Thank you for your feedback and accepting to use measures of heterogeneity as

manuscript, I recommend tau instead of i^2 and chi^2, as tau is a direct measure of heterogeneity rather than indirect. But I understand the authors wanted to follow the Cochrane recommendation.	recommended by Cochrane
4. The methods sections states "We interpreted the certainty of evidence according to guidance provided by the GRADE working group, which takes into consideration the size of the effect and the certainty of evidence." You seem reluctant to be transparent about the implicit numerical thresholds you used when saying "little or no effect" vs "very small effect" vs "may increase". The GRADE guidance recommends transparency. At the very least please change "We interpreted the certainty of evidence according to guidance..." to "We wrote statements about the evidence (e.g., "little or no effect" vs. "very small effect") according to guidance..."	Thanks for the suggestion; we have re-phrased the sentence as suggested.
5. In the new table 3, it seems you made something harder to understand, not easier. Specifically, instead of "probability of CD4 count control was 6% greater in intervention clinics compared to control clinics" you merely have the beta with its CI. I doubt readers will know what that beta means, so i'd prefer the earlier version. Maybe keep the old text and add beta.	Thanks for feedback; we have kept the old version as recommended.
6. For access to care the table only says "There was no change in trend..." And then to the right "little or no effect". However, the left column text did not even provide a point estimate of the effect, thus the reader cannot judge whether it is appropriate to say, "little or no effect". The text below does give that, so i think the table needs the numbers too.	Thanks for highlighting this issue. It is indeed our intention to make the manuscript as comprehensible as possible. Table 3 gives an overview of the evidence that was obtained during this review while more detailed information is reported narratively. We feel that adding the numbers from the text to the table and thus reporting the same information two times in the exact similar format might decrease the overview character of the table. After reading the table readers that are interested in the detailed results will be able to access and assess the effect by what is reported narratively.
In the strategies to promote table, for promoting you wrote "The mean change in HbA1c with integrated care was 0.11 % higher (0.2 lower to 0.42 higher)". Please check the accuracy of that. It sounds like on average patients got slightly WORSE? Whereas the usual care group got slightly better. So, you'll need something in the relative comparison. column, which it seems will at least slightly favor usual care. These are two RCTs so probably they reported enough info for you to calculate a relative effect size. Same point	We understand your concern. The outcome looks at change in HbA1c and not HbA1c. Our results mean that the effect estimate shows a less pronounced change in the intervention group as compared to the control group. To make this clearer we added this detail to the narrative description of the results. It now reads: “Results from two cluster RCTs suggest that interventions to promote integrated care compared to usual

with SPB (promotion group got a little worse but the usual care group got a little better).	care may have little or no effect on change in HbA1c from baseline to follow-up (MD 0.11%; 95%CI -0.20 to 0.42; n=1687; 2 RCTs, low-certainty evidence). This means that the change in HbA1c was similar in both groups.”
In the strategies to promote table, for changes in total cholesterol, you wrote "The mean change in total cholesterol with usual care was 2.0 mg/dl" but is that an increase or a decrease? I'd also like to see numbers in the relative column, in order to judge the appropriateness of your statement that there was "little or no effect"	We gladly corrected the text to indicate the increase in mean cholesterol in the control group as reported by the study authors. The table now says: “The mean change in total cholesterol with usual care was 2.0 mg/dl higher” Usually, GRADE Summary of findings tables do only report the absolute effects for continuous outcomes. The difference between the groups is depicted in the absolute effect that says that the change was 2.5 mg/dl lower as compared to the control group and the 95% CI reports the statistical uncertainty around that effect.